

home: 7.8 km

HOME

Shah-rahe Sher Shah Suri Road
NORTH NAZIMABAD

Allama Rasheed Torbani Road

Shah-rahe Paliistan

7th St.

Masoomen Road

Shah Jahany Road

Ali Murtuza Road

Allama Shabbir Ahmael Usmani Road

University Road

UNIVERSITY

GULSHAN-E-IBBAL

Shah-rahe Anwarzeb

sakhi hassan

people chorangi

water pump

Rizwan park

sorab goth

nip: 4km

gulshan ch: 3.2km

luckypore ch: 5.5km

sorab goth: 8 km

rizwan park: 5km

water pump: 5.2 km

people ch: 6km

sakhi hassan: 9km

nagin ch: 10km

continental: 3km

Shah-rahe Anwarzeb
Rashid Road
Luckypore chorangi
gulshan chorangi
nipa

five star

landi kotal

aysha manzil

mukka chowk

zia colony

• five star: 7.5 km

• landi kotal: 5.9 km

• aysha manzil: 5 km

• mukka chowk: 4.8 km

• zia colony: 4 km

str 1

str 2

str 3

str 4

continental bakers

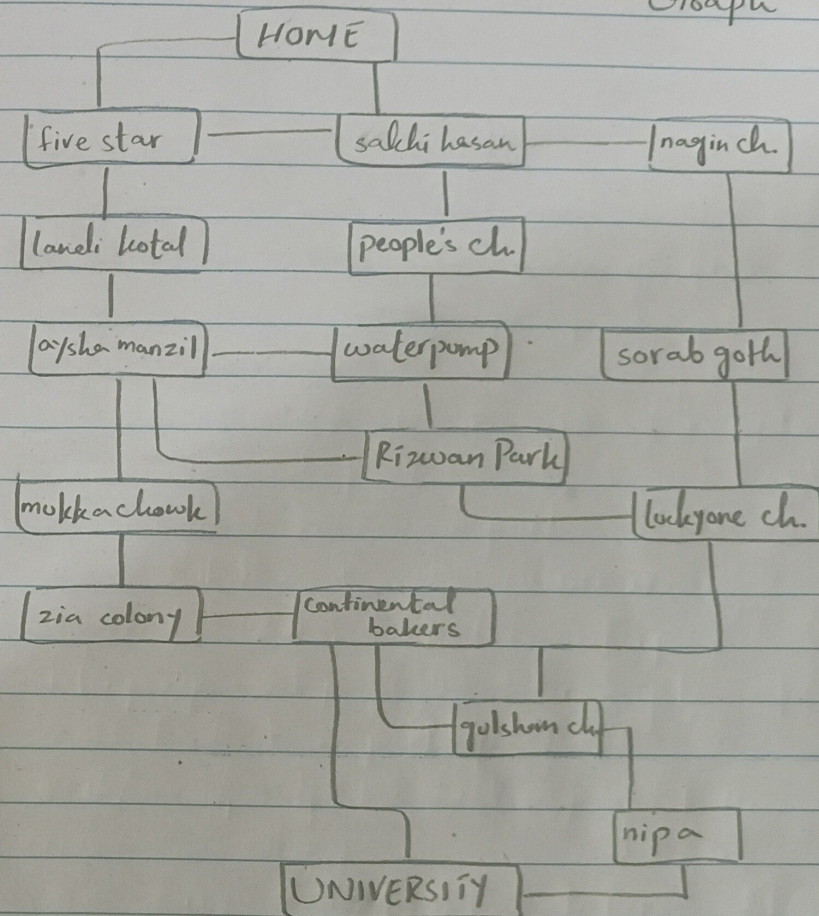
Transition Table

- HOME → five star, sahehi hasan
- five star → sahehi hasan, home, landeli kotal
- sahehi hasan → five star, home, people's ch.,
nagin ch.
- nagin ch. → sahehi hasan, sorab goth
- landeli kotal → people ch., five star, aysha
manzil.
- aysha manzil → Rizwan park, mukka chowk,
landeli kotal.
- people ch. → landeli kotal, waterpump, sahehi hasan
- waterpump → people's ch., sorab goth, aysha
manzil, Rizwan park
- mukka chowk → aysha manzil, Rizwan
Park, zia colony
- zia colony → continental bakers, mukka
chowk.
- Continental bakers → gulshan ch., zia colony,
University
- sorab goth → nagin ch., waterpump, lucky one
ch.,
- Rizwan park → waterpump, aysha manzil,

mokken chowk, luckyone ch.

- luckyone ch → gulshan ch., sorab goth, Rizwan park
- gulshan ch → continental bakers, nipa, luckyone
- nipa → University, gulshan ch.
- University → nipa, continental bakers

Graph



①

A* Algo:-

Problem Home to Uni.

Home $\rightarrow$ None

PQ: [Home] V: [Home]

V: [Home]

Initialization:- $\therefore f(n) = h(n) + g(n)$

State	$g(n)$	$h(n)$	$f(n)$
- Home	0	7.8	7.8

• Visit Home, Neighbours: [five star, sakhi hasan]

PQ: [five star, sakhi hasan]

V: [Home, five star, sakhi hasan]

five star	1	7.5	8.5
sakhi hasan	1	9	10

Parent Table

Home $\rightarrow$ None, five star $\rightarrow$ Home

(2)

- Visit five star:

Neighbours: salhi hasan, laneli kotal, Home

- laneli kotal 2 5.9 7.9

PQ: [salhi hasan, laneli kotal]

V: [Home, five star, salhi hasan, laneli kotal]

Parent Table:

Home → None, five star → Home

laneli kotal → five star

- Visit laneli kotal

Neighbours: people's cherangi, aysa manzil

PQ: [salhi hasan, aysa manzil, people ch.]

V: [Home, five star, salhi hasan, laneli kotal, people's ch., aysa manzil]

- people's ch. 3 6 9

- aysa manzil 3 5 8

(3)

Parent Table:

Home → None, five star → Home

laneli kotal → five star

aysa manzil → laneli kotal

- Visit Aysa manzil

Neighbours: laneli kotal, waterpump, Rizwan Park, mukka chowk

PQ: [salhi hasan, people ch., waterpump, Rizwan Park, mukka chowk]

V: [Home, five star, salhi hasan, laneli kotal, people's ch., aysa manzil, waterpump, Rizwan Park, Mukka chowk]

waterpump 4 5.2 9.2

- mukka chowk 4 5 9

Rizwan Park 4 5.1 9.1

Parent Table

Home → None, five star → Home

laneli kotal → five star

aysa manzil → laneli kotal

mukka chowk → aysa manzil

(4)

• Visit mukka chowk

Neighbours: aysa manzil, zia colony, Rizwan Park

PQ: [sakhi hasan, people ch., water pump, Rizwan, zia]

V: [Home, five star, sakhi hasan, laneli kotal, people ch., aysa manzil, water pump, Rizwan Park, Mukka chowk, zia colony]

zia colony

5

4

9

Parent Table

Home → None, five star → Home

laneli kotal → five star

aysa manzil → laneli kotal

mukka chowk → aysa manzil

zia colony → mukka chowk

• Visit Zia colony

Neighbours: mukka chowk, continental bakers

PQ: [sakhi hasan, people ch., water pump, Rizwan, zia, continental bakers]

5
V: [Home, five star, sakhi hasan, people ch.,
laneli kotal, ayshe manzil, waterpump, Rizwan
Park, Mukka chowk, zia colony, continental
bakers]

continental bakers	6	3	9
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Parent Table

Home → None , five star → Home

laneli kotal → five star

ayshe manzil → laneli kotal

mukka chowk → ayshe manzil

zia colony → mukka chowk

continental → zia colony

• Visit Continental:

Neighbours: University, golshan ch.

University	1	0	7
golshan ch.	7	3-2	10-2

PQ: [sakhi hasan, people ch., water pump,
Rizwan Park, University, golshan ch.]

V: [Home, five star, sakhi hasan, laneli
kotal, people ch., aysa manzil, water
pump, Rizwan Park, Mukka chowk,
zia colony, Continental bakers, golshan
ch., University]

Parent Table

Home → None, five star → Home

laneli kotal → five star

aysa manzil → laneli kotal

mukka chowk → aysa manzil

zia colony → mukka chowk

continental → zia colony

University → continental

University → continental → zia colony →
mukka chowk → aysa manzil → laneli
kotal → Home.

Home → laneli kotal → aysa manzil →

mukka chowk → zia colony → continental
→ University.